

AYE
FIT HUB

• MND • DAY 12

Sleep & Recovery

The Overlooked Performance Tool

Day 12 of 30 | Mindset & Recovery

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- Why 8 hours of bad sleep erases your workout gains
- The 5 sleep stages and what each one repairs
- How cortisol, HGH and testosterone are ruled by sleep
- The 10 science-backed sleep hygiene habits
- Active recovery protocols for Day 12 rest day

01 | Why Sleep is Your #1 Performance Drug

Every training adaptation — muscle growth, fat loss, strength gains, cardiovascular improvement — happens **while you sleep**, not while you train. Exercise is the stimulus; sleep is where the body responds. You can eat perfectly and train hard six days a week, but if you sleep five hours a night you are leaving 40–60% of your results on the table.

Research from the University of Chicago found that **sleep-deprived dieters lost 55% less fat and 60% more muscle** than those who slept 8.5 hours. The same training programme produced fundamentally different outcomes based purely on sleep duration.

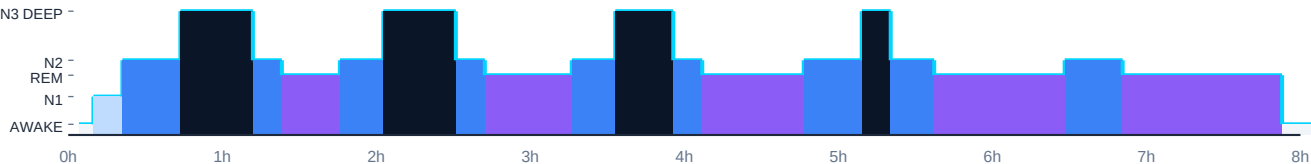
The Sleep-Performance Equation

Sleep deprived (< 6 hrs): Testosterone –10–15%, HGH –70%, cortisol +37%, reaction time = legally drunk. Well rested (7–9 hrs): Peak anabolic hormone output, full glycogen resynthesis, complete neural repair, optimised fat metabolism.

02 | The 5 Sleep Stages — What Gets Repaired

Sleep is not a single state — it cycles through five distinct stages across 90-minute cycles, repeated 4–6 times per night. Each stage performs different repair and consolidation functions.

Figure 1 — Sleep Architecture: Hypnogram (8-Hour Night)



Stage	Name	Duration/Night	What Gets Repaired
N1	Light Sleep	5% (~25 min)	Transition to sleep; muscle relaxation begins; easily woken
N2	True Sleep	45% (~3.5 hrs)	Core body temp drops; heart rate slows; memory consolidation starts
N3	Deep / Slow-Wave Sleep	15–25% (~1.5 hrs)	HGH release (80% of daily total); muscle protein synthesis; immune repair; tissue rebuilding
REM	Rapid Eye Movement	20–25% (~1.5 hrs)	Motor skill consolidation; mental fatigue recovery; emotional regulation; dreaming
Brief Wake	Micro-arousals	<5%	Normal — unnoticed arousals between cycles; increases with age

03 | Sleep, Hormones and Body Composition

Your hormonal environment is almost entirely governed by your sleep schedule. The three most important hormones for athletes and body composition — HGH, testosterone, and cortisol — all follow circadian patterns that are disrupted by poor sleep.

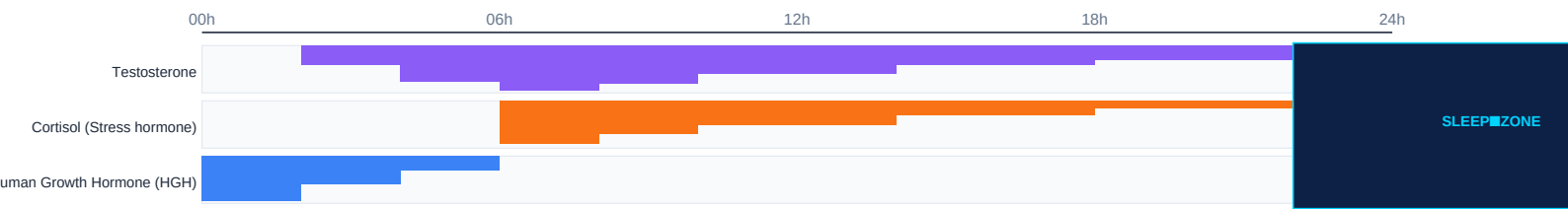


Figure 2 — Key Hormone Levels Across 24 Hours (Sleep Zone Shaded)

Hormone	Sleep Effect	Consequences of Sleep Loss
Human Growth Hormone (HGH)	Peaks in first 2 hrs of sleep (N3 stage). 80% of daily HGH is released during deep sleep.	Reduced muscle protein synthesis, slower fat metabolism, impaired tissue repair, faster biological ageing
Testosterone	Peaks during REM sleep in early morning hours (5–7 am). Requires 7+ hrs for full production.	< 6 hrs sleep → testosterone equivalent to ageing 10–15 years. Lower libido, reduced strength, reduced motivation
Cortisol (Stress hormone)	Naturally low during sleep; rises sharply at 6–8 am to promote waking. Healthy spike = energy. Chronic elevation = catabolic.	Sleep deprivation keeps cortisol elevated all day → muscle breakdown, fat storage (especially abdominal), impaired immunity, anxiety
Leptin & Ghrelin (Appetite hormones)	Leptin (satiety) rises during sleep. Ghrelin (hunger) suppressed with adequate sleep.	Sleep loss → leptin –18%, ghrelin +28% → 300–500 extra calories consumed next day; cravings for high-carb, high-fat foods

The Cortisol Trap

Late-night training (after 9 pm), screen light, caffeine after 2 pm, and alcohol all elevate cortisol and suppress melatonin — the sleep hormone. One glass of alcohol reduces deep sleep by 20% even if it helps you fall asleep faster.

04 | 10 Science-Backed Sleep Hygiene Habits

#	Habit	The Science
1	Consistent sleep/wake time (±30 min)	Anchors your circadian rhythm; irregular sleep is as damaging as short sleep
2	Keep bedroom 16–19°C	Core body temp must drop 1–2°C to initiate sleep; cooler room accelerates this
3	Complete darkness (blackout curtains)	Even small light exposure suppresses melatonin by 50%; covers eyes with mask if needed
4	No screens 60 min before bed	Blue light from phones/TV delays melatonin by 2–3 hrs; use night mode minimum
5	No caffeine after 2 pm	Caffeine half-life = 5–7 hrs; a 3 pm coffee is still 50% active at 9 pm
6	No alcohol within 3 hrs of sleep	Alcohol fragments sleep architecture, suppresses REM and deep sleep
7	Wind-down routine (20–30 min)	Shower, light reading, breathing exercises signal nervous system to downshift
8	Morning sunlight (10–20 min)	Sets circadian clock; triggers cortisol wake spike at right time; improves night sleep
9	Magnesium glycinate (200–400 mg)	Most people are deficient; improves sleep quality and reduces nocturnal cortisol
10	Reserve bed for sleep only	Psychological association: bed = sleep cue. Working/scrolling in bed breaks this

05 | Day 12 Recovery Protocol

Day 12 is a designated recovery day. That does not mean do nothing — it means do the **right** things that accelerate adaptation from the last 11 days of training. Active recovery keeps blood moving, reduces soreness, and improves tomorrow's performance.



Figure 3 — Recovery Priority Pyramid

Recovery Method	Duration	Benefit	When
20-min walk (low intensity)	20 min	Increases blood flow to muscles; clears lactate; reduces DOMS	Morning or afternoon
Full-body stretch / yoga flow	15–20 min	Restores range of motion; parasympathetic activation	After walk or evening
Cold shower (last 60–90 sec cold)	90 sec	Reduces inflammation; boosts alertness; improves mood (dopamine +250%)	Morning

Foam rolling — tight areas	10 min	Breaks up fascial adhesions; reduces perceived soreness next day	Evening
Hydration: 2.5–3 L water	All day	Plasma volume supports muscle repair; flushes metabolic waste	Throughout day
Sleep 7.5–9 hrs tonight	7.5–9 hrs	Deep sleep HGH pulse; full REM cycle; hormonal reset	10 pm – 6 am target

Day 12 Action

Tonight: set a consistent bedtime and remove your phone from your bedroom. Tomorrow you will train again — the quality of that session depends entirely on the recovery you do tonight. Next lesson — Day 13: Upper Body Power — Push, Pull, Press.